Academic Collaboration Network: A Systems Engineering Approach to Mitigate Isolated Learning Through Skill-Based Group Optimization

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Agenda

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Problem Statement & Limitations

Current collaboration challenges in higher education

02

Methodology

Research approach and analytical framework

03

System Architecture & Mathematical Formulation

Technical design and optimization models

04

Results: Survey & Simulation

Empirical findings and performance metrics

05

Conclusions & Future Work

Key achievements and research extensions

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Problem Statement & Limitations

Survey Findings

88%

WhatsApp Usage

Students rely exclusively on WhatsApp for group coordination

36%

Communication Barrier

Primary obstacle to effective collaboration

96%

Platform Interest

Express willingness to adopt dedicated system

Systemic Limitations

1. Lack of structured group formation protocols
2. Skill clustering concentrates high performers
3. Student isolation impedes knowledge sharing
4. Inefficient knowledge exchange mechanisms
5. Absence of skill visibility across cohort

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Methodology Overview

Survey

7 questions on collaboration patterns, Google Forms, n=25 respondents

Systems Analysis

Stakeholder identification, causal loop analysis, sensitivity evaluation

Simulation

Synthetic profiles, normal distribution (μ=0.5, σ=0.2), 30 trials

Optimization

Complementary variance minimization algorithm

Validation

Metrics comparison against random and similarity-based grouping

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System Architecture

Microservices architecture enabling scalability to 500+ concurrent users with <2s response times and 99.5% availability

API Gateway

Core Services

- Auth & Authorization
- Profile Service
- Matching Engine
- Workspace
- Notification
- Analytics
- Integration Gateway

Data Layer

- PostgreSQL
- Redis
- S3

External Integrations

- LMS (Moodle)
- Library System

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Mathematical Formulation

Student Profile Vector

$$F_i = \{L_i, A_i, P_i, H_i\}$$

Multi-dimensional skill representation capturing learning capacity, analytical ability, programming proficiency, and hard skills

Optimization Objective

$$\min \sum \mathrm{Var}(L_g) \text{ subject to } |G_g| = m$$

Minimize intra-group variance while maintaining group size constraints

Group Balance Index

$$\mathrm{GBI} = 1 - \frac{1}{k} \sum \frac{\mathrm{Var}(L_g)}{\mathrm{Var}_{\mathrm{total}}}$$

Normalized measure of skill distribution balance across all groups (0-1 scale)

Network Metrics

$$C_D(v_i) = \frac{\deg(v_i)}{n-1}$$

$$\mathrm{Density} = \frac{2|E|}{|V|(|V|-1)}$$

Centrality measures and network connectivity analysis

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Survey Results

Main Barriers to Group Formation

Communication breakdowns represent the primary obstacle to effective study group formation

Adoption Likelihood Distribution

Mean score of 3.88/5 indicates strong adoption意愿

Key Insight: 96% of students express interest in a dedicated platform for academic collaboration

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Simulation Results

Metric	Random	Similarity	Complementary
Skill Variance	0.142	0.089	**0.041 ↓71%**
Network Density	0.23	0.31	**0.42 ↑83%**
Centrality Gini	0.38	0.29	**0.17 ↓55%**
GBI	0.51	0.67	**0.85 ↑67%**

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Conclusions

71%

Skill Dispersion

Reduction in intra-group variance

83%

Network Density

Improvement in connectivity

55%

Centrality Fairness

Enhanced equity distribution

96%

Platform Interest

Validated adoption意愿

500+ Concurrent Users

<2s Response Times

99.5% Availability

> "The systems engineering approach effectively transforms informal collaboration into structured, measurable processes"

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Future Work & Q&A

- Machine learning-based predictive grouping
- Dynamic reconfiguration during semester progression
- Integration with LMS and SIA systems
- Real-world pilot implementation
- Behavioral compatibility factors

Repository: https://github.com/AnveloCZ/Academic-Collaboration-Network-SAD-2026-1

Thank you

Questions?

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